Chhiring Analysis

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Contents

Import and clean the dataset

```
df <- read.xlsx("Proj_4_data.xlsx") |>
  janitor::clean_names() |>
  mutate(time = case_when(time == 1 ~ "Baseline",
                          time == 2 ~ "6 Month",
                         time == 3 ~ "12 Month"),
         time = fct_relevel(time, "Baseline", "6 Month"),
         group = case_when(group == 1 ~ "Asthma-PASS", ##intervention
                           group == 0 ~ "AM"),  ##standard of care
         group = as.factor(group),
         school = as.factor(school)) |>
  as_tibble()
```

324 total observations, has 32 observations with 0 –> sparse?

108 people across (49 control:21,28 in each site, 59 tx: 28, 31 in each site)

###Summary Table (observations per time point)

```
table1a  <- df %>%
  drop_na() |>
  mutate(`Baseline SFD` = case_when(time == "Baseline" ~ as.integer(sfd),
                                    time != "Baseline" ~ NA)) |>
  dplyr::select(time, school, group, `Baseline SFD`) %>%
  tbl_strata(
      strata = group,
      .tbl_fun = ~ {
        dat <- .x %>%
          dplyr::mutate(
            school = factor(as.character(school))
          )

        tbl_summary(
          dat,
          by = school,
          missing = "no",
          label = list(
            time = "Time"
```

```
          )
        ) %>%
          add_p() %>%
          add_overall() |>
          bold_labels()
      },
      .header = "**{strata}**, N = {n}"
    )
```

```
## The following warnings were returned during `tbl_strata()`:
## ! For variable `Baseline SFD` (`school`) and "estimate", "statistic",
##   "p.value", "conf.low", and "conf.high" statistics: cannot compute exact
##   p-value with ties
## ! For variable `Baseline SFD` (`school`) and "estimate", "statistic",
##   "p.value", "conf.low", and "conf.high" statistics: cannot compute exact
##   confidence intervals with ties
## The following warnings were returned during `tbl_strata()`:
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## ! For variable `Baseline SFD` (`school`) and "estimate", "statistic",
##   "p.value", "conf.low", and "conf.high" statistics: cannot compute exact
##   confidence intervals with ties
```

```
table1a |>
  as_gt() |>
  tab_options(
    table.font.size = px(12)) |>
  cols_width(
    label      ~ px(120),
    everything() ~ px(70)
  ) |>
  gt::gtsave(filename = "figures_tables/tab1a.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpBoWFWZ/file12536544210c4.html screensh
```

```
table1a |>
  as_kable()
```

Characteristic	**Overall** N = 137	**2** N = 60	**4** N = 77	**p-value**	**Overall** N = 159	**1** N = 75	**3** N = 84	**p-value**
Time				0.9				>0.9
Baseline	48 (35%)	21 (35%)	27 (35%)		57 (36%)	27 (36%)	30 (36%)	
6 Month	46 (34%)	19 (32%)	27 (35%)		52 (33%)	25 (33%)	27 (32%)	
12 Month	43 (31%)	20 (33%)	23 (30%)		50 (31%)	23 (31%)	27 (32%)	
Baseline SFD	11 (8, 14)	11 (9, 14)	11 (7, 13)	>0.9	11 (7, 12)	12 (7, 13)	9 (6, 12)	0.11

```
table1b <-  df %>%
  drop_na() |>
  mutate(`Baseline SFD` = case_when(time == "Baseline" ~ as.integer(sfd),
                                    time != "Baseline" ~ NA)) |>
  dplyr::select(time, group, `Baseline SFD`) |>
  tbl_summary(by = group,
              label = list(time = "Time"),
              missing = "no") |>
  add_overall() |>
  add_p() |>
  modify_spanning_header(all_stat_cols() ~ "**Overall**")

table1b |>
  as_gt() |>
  tab_options(
    table.font.size = px(12)) |>
  cols_width(
    label      ~ px(120),
    everything() ~ px(80)
  ) |>
  gt::gtsave(filename = "figures_tables/summarytab.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpBoWFWZ/file125367caaae81.html screensho
```

```
table1b |>
  as_kable()
```

Characteristic	**Overall** N = 296	**AM** N = 137	**Asthma-PASS** N = 159	**p-value**
Time				>0.9
Baseline	105 (35%)	48 (35%)	57 (36%)	
6 Month	98 (33%)	46 (34%)	52 (33%)	
12 Month	93 (31%)	43 (31%)	50 (31%)	
Baseline SFD	11 (7, 13)	11 (8, 14)	11 (7, 12)	0.4

##N_irt Calculation

```
sig = 0.05/4
power = 0.80
d = 1/3
z_1minusa = qnorm(1-sig/2)
z_power = qnorm(power)
n_pergroup = 2*(z_1minusa + z_power)^2/(d^2)
n_pergroup
```

```
## [1] 200.7199
```

model from pilot study:

```r
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.5.2

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.5.2

##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```r
data_grouped <- df %>%
  mutate(sfd = sfd,
         no_sfd = 14 - sfd)

# mod_num <- glmer(
#   cbind(sfd, no_sfd) ~ time * group + (1 | school/id),
#   family = betabinomial,
#   data = data_grouped,
#   control = glmerControl(optimizer = "bobyqa",
#                          optCtrl = list(maxfun = 2e5))
# )

library(glmmTMB)
```

```
## Warning: package 'glmmTMB' was built under R version 4.5.2
```

```r
mod_num <- glmmTMB(cbind(sfd, no_sfd) ~ time * group + (1 | school/id),
                   data = data_grouped,
                   family = betabinomial)

##sigma_b
```

```
sigma_b = attr(VarCorr(mod_num)$cond$`id:school`, "stddev")
sigma_c = attr(VarCorr(mod_num)$cond$`school`, "stddev")
names(sigma_b) = NULL
names(sigma_c) = NULL
##resiudal
sigma = sd(residuals(mod_num))
#sigma = pi^2/3
```

```
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.2

## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
emm <- emmeans(mod_num, ~ group | time)
cont = contrast(emm, method = "revpairwise")
summary(cont, infer = c(TRUE, TRUE), side = ">", adjust = "bonferonni")
```

```
## time = Baseline:
##  contrast           estimate    SE  df asymp.LCL asymp.UCL z.ratio p.value
##  (Asthma-PASS) - AM   -0.289 0.339 Inf    -0.846       Inf  -0.852  0.8028
##
## time = 6 Month:
##  contrast           estimate    SE  df asymp.LCL asymp.UCL z.ratio p.value
##  (Asthma-PASS) - AM    0.226 0.380 Inf    -0.399       Inf   0.594  0.2761
##
## time = 12 Month:
##  contrast           estimate    SE  df asymp.LCL asymp.UCL z.ratio p.value
##  (Asthma-PASS) - AM    0.209 0.377 Inf    -0.411       Inf   0.554  0.2900
##
## Results are given on the log odds ratio (not the response) scale.
## Confidence level used: 0.95
## P values are right-tailed
```

Ncrt accounting for ICC (previous version)

```
rho = sigma_c^2/(sigma_b^2 + sigma_c^2 + sigma^2) #(ICC) need to replace once we have this info from Ka
r = -0.044 #(based on baseline to 6 month and baseline to 12 month correlation)
k = 30/2 #15 schools per arm
n_irt_total = n_pergroup
m = (n_irt_total*(1-rho))/(k - n_irt_total*rho)
```

NCrt with assumed ICC (previous version)

```
sig = 0.05/4
power = 0.80
d = 1/3
z_1minusa = qnorm(1-sig/2)
z_power = qnorm(power)
n_pergroup = 2*(z_1minusa + z_power)^2/(d^2)
n_pergroup
```

```
## [1] 200.7199
```

```
icc_list <- c(0, 0.01, 0.02, 0.03, 0.04, 0.05)

get_m <- lapply(icc_list, function(rho){
  print(rho)
  k = 30/2 #15 schools per arm
  n_irt_total = n_pergroup
  n_rho = n_irt_total*rho
  m = (n_irt_total*(1-rho))/(k - n_irt_total*rho)
  n = ceiling(m)*k
  m_adj = m/(1-0.2)
  n_adj = ceiling(m_adj)*k
  df <- as_tibble(
    list(Nirt_rho = round(n_rho, digits =3),
         M = round(m, digits =3),
         N = n, M_adj = round(m_adj, digits =3), N_adj = n_adj)
  )
  return(df)
})
```

```
## [1] 0
## [1] 0.01
## [1] 0.02
## [1] 0.03
## [1] 0.04
## [1] 0.05
```

```
assumed_icc_samp <- Reduce(rbind, get_m)
```

NCrt (using before and after observations)

```
rho = sigma_c^2/(sigma_b^2 + sigma_c^2 + sigma^2) #(ICC) need to replace once we have this info from Ka
r = -0.044 #(based on baseline to 6 month and baseline to 12 month correlation)
k = 30/2 #15 schools per arm

a = -2*rho*k
b = 4*n_irt_total*rho*(1-rho) + 2*k*(rho -1)
c = 2*n_irt_total*(1 + rho^2) - 4*n_irt_total*rho
mo = (-b - sqrt(b^2 -4*a*c))/(2*a)
m1 = (-b + sqrt(b^2 -4*a*c))/(2*a)
m = c(mo, m1)
m = m[m > 0]
```

NCrt with assumed ICC

```
sig = 0.05/4
power = 0.80
d = 1/3
z_1minusa = qnorm(1-sig/2)
z_power = qnorm(power)
n_pergroup = 2*(z_1minusa + z_power)^2/(d^2)
n_pergroup
```

```
## [1] 200.7199
```

```
rho = sigma_c^2/(sigma_b^2 + sigma_c^2 + sigma^2) #(ICC) need to replace once we have this info from Ka
r = -0.044 #(based on baseline to 6 month and baseline to 12 month correlation), no need since assumed
k = 30/2 #15 schools per arm


icc_list <- c(0.005, 0.01, 0.02, 0.03, 0.04, 0.05)

get_m <- lapply(icc_list, function(rho){
  print(rho)
  k = 30/2 #15 schools per arm
  n_irt_total = n_pergroup
  n_rho = n_irt_total*rho

  a = -2*rho*k
  b = 4*n_irt_total*rho*(1-rho) + 2*k*(rho -1)
  c = 2*n_irt_total*(1 + rho^2) - 4*n_irt_total*rho
  mo = (-b - sqrt(b^2 -4*a*c))/(2*a)
  m1 = (-b + sqrt(b^2 -4*a*c))/(2*a)
  m = c(mo, m1)
  print(m)
  m = m[m > 0]

  n = ceiling(m)*k
  m_adj = m/(1-0.2)
  n_adj = ceiling(m_adj)*k
  df <- as_tibble(
    list(Nirt_rho = round(n_rho, digits =3),
         M = round(m, digits =3),
         N = n, M_adj = round(m_adj, digits =3), N_adj = n_adj)
  )
  return(df)
})
```

```
## [1] 0.005
## [1]   14.20129 -186.57245
## [1] 0.01
## [1]  14.98955 -87.49453
## [1] 0.02
## [1]  16.40253 -39.17513
## [1] 0.03
## [1]  17.54576 -23.91933
## [1] 0.04
## [1]  18.42508 -16.73294
## [1] 0.05
## [1]  19.08209 -12.65757
```

```
assumed_icc_samp <- Reduce(rbind, get_m)
```